Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CRISISLTLSUM* | ***Context:*** *Ecuador — Humanitarian Crisis Social Media Analysts*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All 10,610 tweets in the benchmark were collected from seven US states and two Australian states during 2020–2021 [Q123], selected explicitly because their event profiles match the benchmark's four crisis categories [Q124]. Tweet filtering relies on location candidates from US/Australian cities, towns, and 'famous neighborhoods' [Q30] and on English-curated keyword lists [Q126, Q127]. No Spanish-language, Latin American, indigenous-language, or humanitarian-organizational content is documented. The Ecuador deployment requires anchoring to parroquias, cantones, provincias, SNGR/SGR (renamed in 2024 [WEB-11]), Cruz Roja Ecuatoriana, IGEPN, INAMHI, and Kichwa/Shuar toponyms — none of which are present. The elicitation confirms US-default references would be 'operationally meaningless or misleading.' This is an instance-level content-validity violation that cannot be remediated through fine-tuning alone.

ID	Evidence
Q123	“We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria.” (p.12)
Q124	“These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York.” (p.13)
Q30	“The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest.” (p.3)
Q126	“We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.” (p.13)
Q127	“Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm".” (p.13)
WEB-11	National disaster authority renamed from SGR to SNGR in 2023–2024 (https://www.gestionderiesgos.gob.ec/wp-content/uploads/downloads/2024/05/Resol.SNGR-102-2024.pdf)

Strengths

- The benchmark's clustering and filtering methodology (geographic+keyword+entity-based clustering) is documented in sufficient detail [Q28, Q30, Q31] to be re-implemented for Ecuador, providing a useful methodological template even though no instance content transfers.

ID	Evidence
Q28	“Figure 2 shows the process for generating a set of noisy timelines starting from the Twitter stream and followed by pre-processing and knowledge enhancement steps, the online clustering method, and post-processing & cleaning steps.” (p.3)
Q30	“The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest.” (p.3)

Q31	“To limit the tweets to a specified crisis domain, we curate domain-specific keywords and only select tweets with phrases matching one of the keywords. This approach is not comprehensive or exhaustive but somewhat representative of each crisis domain.” (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Ecuador deployment requires Spanish-language content, Ecuadorian administrative geography (24 provinces [WEB-6], ~221–224 cantons [WEB-8, WEB-9], 1,500+ parroquias [WEB-8]), Ecuadorian institutional handles (SNGR/SGR [WEB-11], IGEPN, INAMHI, Cruz Roja Ecuatoriana), and Kichwa/Shuar place names. None are present in the benchmark.

IC-2: Culturally sensitive content (e.g., 'damnificados', 'albergue', references to indigenous communities, displacement vocabulary) is entirely absent. The benchmark contains no Latin American or humanitarian-organizational instances.

IC-3: All instances require US/Australian-specific knowledge: place names from California, Texas, Florida, New York, Illinois, Massachusetts, NSW, Victoria [Q123]; English-language emergency vocabulary; US/Australian agency references implicit in keyword lists [Q126].

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator review of benchmark instances has been performed; the elicitation confirms an inter-operational team of Ecuadorian field coordinators would need to do this.

IC-5: Documented: 100% of input instances are US/Australian English Twitter content [Q123]; transferability to Ecuadorian Spanish humanitarian context is essentially zero at the instance level.

ID	Evidence
Q123	“We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria.” (p.12)
WEB-6	Ecuador has 24 provinces (Pichincha, Guayas, Manabí, etc.) (https://en.wikipedia.org/wiki/Provinces_of_Ecuador)
WEB-8	Ecuador has approximately 221 cantons and over 1,500 parroquias (https://storyteller.travel/map-provinces-ecuador/)
WEB-11	National disaster authority renamed from SGR to SNGR in 2023–2024 (https://www.gestionderiesgos.gob.ec/wp-content/uploads/downloads/2024/05/Resol.SNGR-102-2024.pdf)
WEB-9	https://en.wikipedia.org/wiki/Cantons_of_Ecuador
Q126	“We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.” (p.13)
Q127	“Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm".” (p.13)
Q30	“The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest.” (p.3)
Q69	“Out of these 1,000 annotated timelines (10,610 tweets) in CrisisLTLSum, 423 (42%) are about” (p.5)
WEB-17	UrbangEnCy provides Ecuador-specific Spanish emergency tweets but only binary classification (https://www.sciencedirect.com/science/article/pii/S2352340920315729)

Information Gaps

- The proportion of operationally critical Ecuador crisis tweets in Kichwa or with Kichwa toponyms is not quantified in available sources.

Requires Expert Verification

- Authoritative list of Ecuadorian institutional handles and parroquia-level toponyms; should be elicited from CONAIE, ECUARUNARI, SNGR, and Cruz Roja Ecuatoriana.

Remediation

Gap 1: All instances are US/Australian English Twitter from 2020–2021; no Ecuadorian Spanish, Kichwa toponym, or humanitarian-organization content is present.

Recommendation: Build a parallel Ecuadorian Spanish crisis timeline corpus by extending UrbangEnCy [WEB-17] with timeline structure and adding seed clusters for volcanic, seismic, and displacement events; curate parroquia/cantón/Kichwa-toponym lists with CONAIE/ECUARUNARI input.

ID	Evidence
WEB-17	UrbangEnCy provides Ecuador-specific Spanish emergency tweets but only binary classification (https://www.sciencedirect.com/science/article/pii/S2352340920315729)